

Sitting With Data: Data Weaving as a Communal Practice

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Abstract

Craft-based approaches to working with data are increasingly explored; however they are rarely explored as an ongoing practice of different communities. In this paper, we introduce data weaving as a collective practice of working with data through bead-woven bracelets. To explore how this practice unfolds, we conducted a qualitative study of a self-organized community, based on a group interview with four people of the community. We focus on how data decisions are made in situ, as participants adapt datasets to material constraints, respond to changing or incomplete data, and discuss questions of privacy and public display. Our findings show that crafting with data shifts data work toward shared interpretation. We contribute with data weaving as a communal form of sensemaking through data that foreground process, relationships and aspects of care among people who perform it.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**.

Keywords

Data Weaving, Data Physicalization, data beads, craft, VisArt, data practices

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1 Introduction

Craft-based approaches to working with data have shown how making with physical materials can create an alternative way of engaging with data [8, 12, 14]. Much of this research centers on individual experiences [14], short-term workshops [10], or the properties of finished artifacts [12], while the longer-term craft practices within ongoing communities remains underexplored.

In this paper, we study a self-organized community that meets regularly to make data-driven craft objects together. We use the term data weaving to describe a collective, long-term practice of working with data through weaving-based craft techniques. Unlike prior work that centers outcomes, individual aspects, or artifact

properties, our account foregrounds the social infrastructure of the practice. Although data weaving appears in rhetoric scholarship for "textiles as data materialization" [7], we use the term to name the community practice of weaving data. Drawing on a group interview, we explore how data weaving operates as a long-term, community-based practice. In doing so, we offer data weaving as a way to think about what long-term, community-based making with data can look like and hope it opens discussions about how shared craft practices might broaden our ideas of data engagement beyond individual experiences and short-term interventions.

2 Background

Previous works present making as a social and political activity shaped by values, identities, and relations, rather than only as a technical process [13]. Taylor et al. [13] describe making as a political act through which people intervene in existing conditions and imagine alternatives. Studies of makerspaces further show how infrastructures and organizational choices shape participation, influence whose knowledge is recognized, and determine which practices are supported. While early work focused on hobbyists and technology enthusiasts, later studies include feminist makers, older adults, stay-at-home fathers, and participants from community-based organizations [13]. In the context of design justice, research shows how making is often associated with technological tinkering, while craft practices are positioned as secondary or informal [9]. Studies of community-led makerspaces for refugee and migrant women document alternative ways of organizing making around care and mutual support, challenging narrow definitions of who counts as a maker and what kinds of work are valued [9]. Related research on craft-based maker cultures describes making as a form of care work, where participants emphasize wellbeing and supporting others rather than individual achievement [15].

A related body of work brings data practices into conversation with craft and community sharing. Nissen and Bowers [11] frame data physicalization as a participatory process of data translation, where meaning develops through material engagement. Thudt et al. [14] introduce data craft to describe the manual making of everyday objects that incorporate personal data, emphasizing reflection. Recent research on data textiles shows how creators map data to textile structures and materials [12], and how working by hand shifts control away from algorithms toward human judgment and material constraints during making [6]. Work with older adults demonstrates how craft-based data physicalization can function as an object language, enabling participants to externalize feelings and attach data narratives to material forms [10]. Workshop-based projects such as Data Murals show how collective crafting supports shared sensemaking around community concerns, with both the



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Figure 1: Data weaving sessions. Left: group working together during a regular meetup. Center: translating data into a pattern. Right: weaving a data bracelet.

act of making and the resulting artifacts creating opportunities for discussion and reflection [2].

While much of the work presented here has focused on workshops, individual artifacts or processes, fewer studies explore how data craft practices are sustained within long lasting communities of practice and how this might shape the community’s understanding about data through crafting. Thus, building on previous work, our study explores data weaving as a collective data craft practice centered on bead-woven bracelets. By exploring these processes within an ongoing community, our work extends existing accounts of data craft by attending to how data craft unfolds collectively over time during making.

3 Method

Our study is based on a group interview with four members of a data crafting collective of six members who have been meeting regularly for over a year (See Figure 1). All participants worked with data in their professional lives, including roles in data journalism, data science, data visualization and social science research. One of the participants is also an author of this paper. To address potential conflicts of interest, this author was not involved in the study design, the facilitation of the interview, or the initial analysis of the interview data. All participants were active members of the same community and had prior experience creating data-driven bead and textile artifacts within this shared practice. We conducted a semi-structured group interview designed to elicit reflective accounts of participants’ making practices and their social dimensions. The interview focused on participants’ experiences of creating physical artifacts that embodied data, their motivations for joining the data weaving sessions, their relationships with data and with one another, and the ongoing meaning and afterlife of the crafted artifacts.

The group format was chosen to support collective reflection and to surface shared understandings as well as moments of divergence. All participants were informed about the purpose of the study and the use of audio recording and transcription, provided voluntary informed consent prior to participation, and were anonymized in the reporting of findings. The interview was audio-recorded and transcribed verbatim. The interview data were analyzed using reflexive thematic analysis [3]. Analysis was iterative with themes developing through repeated engagement with the transcripts.

4 Findings

4.1 Crafting with Data as an Ongoing Practice

Participants described engaging with crafting data as emerging gradually rather than being formally initiated. There was no clear starting point or launch event. As one participant put it, “we just started to come together” (P1). What sustained the practice was not institutional structure, but regularity. The group deliberately committed to a bi-weekly rhythm, which participants described as crucial: “we forced ourselves to gather bi-weekly, which was a regular time” (P1). Knowing “exactly when and where we meet” (P2) helped integrate the sessions into everyday life.

Participation remained intentionally open. Members joined, left, and sometimes returned, which was framed positively as “an open, come and go kind of thing” (P1). Over time, a smaller core group provided continuity, while openness supported accessibility and experimentation. Early sessions were described as “a little chaotic” (P1), with materials scattered and processes unclear. Through repetition, the practice gradually stabilized: “after a while, it just settled” (P1). This organization emerged through shared experience rather than rules or hierarchy. Participants also expressed a desire to keep

the practice voluntary and non-obligatory. As one participant noted, “it’s a fun project for me. I don’t want to make it because I have to” (P3). This balance between commitment and informality shaped how the practice evolved over time.

4.2 Crafting with Data as an Embodied and Slow Engagement

Across the interview, participants contrasted material engagement with data over time with their screen-based professional work. Several described spending much of their time “constantly next to my laptop” (P4) and valued crafting as a way to engage with data “in a physical way, a more tactile way” (P4) as a way to take them away from their screens yet keep engaging with data which they have affinity with but in a more playful way. Others emphasized the importance of “step[ping] out of the computer screen” and doing something physical that still connected to data (P1).

The material nature of crafting introduced slowness and demanded attention. Unlike digital tools, mistakes could not be instantly undone. One participant explained, “it’s not like an algorithm that you just rewrite one line... if you messed up, you have to go back and start it again” (P4). This made the process slower and sometimes frustrating but also encouraged care and focus.

Participants frequently described experiences of flow. One noted the satisfaction of “seeing that it evolves and grows” while crafting (P1), while another described it as “absolutely a flow experience” (P4). Importantly, this flow was often collective (P1, P3). Silence during sessions was described as comfortable, allowing participants to focus without pressure to talk.

4.3 Making Together as Social Infrastructure for Care

Beyond making, data crafting sessions functioned as spaces for social and emotional connection. Participants described how working with their hands made personal conversations feel easier and less exposed. As one participant put it, “it’s easy to share it while you are doing something with your hands” (P4). Attention was split between crafting and conversation, which reduced pressure and allowed sharing to emerge organically.

Over time, this led to feelings of closeness and trust. One participant reflected that the practice “got us closer personally and emotionally and like friendship wise” (P4), while another described it as a way to “warm up our friendship” (P2). Participants shared experiences related to relationships, life transitions, and personal struggles, often after months of simply being present together.

The group was consistently described as “warm,” “welcoming,” and “non-judgmental” (P2, P3, P4). Importantly, care was not an explicit goal of the practice. Instead, it emerged indirectly through repeated co-presence and shared activity. Silence was accepted alongside conversation, allowing participants to engage at their own pace.

4.4 Collective Sensemaking Beyond the Artifact

Data crafting was also described as a collective sensemaking practice in which meaning was negotiated socially. Participants frequently discussed how to adapt data to fit the constraints of bead weaving. As one participant explained, “the data should fit the

bracelet somehow” (P4), which often required aggregation, re-scaling, or re-timing. These transformations were discussed collectively and framed as interpretive choices rather than technical compromises.

Decisions about data validity and privacy were also negotiated collectively. In one instance, a participant discovered that an official data source had updated its values while she was already working with them, prompting questions whether to redo the piece or continue with the earlier version. Similar conversations emerged around personal and sensitive data. P3 shared menstrual cycle data and consulted the group about how, and whether, it should be exhibited publicly in an exhibition. In these cases, questions of reliability, appropriateness, and comfort were discussed together. There was no single correct approach. Instead, the group embraced multiple ways of preparing and visualizing data, shaped by both data constraints and personal meaning. Learning occurred informally through observation and mutual help. Early sessions involved significant care labor, with organizers “just helping others and showing how to do” things (P2). Over time, participants developed a shared understanding of materials and methods, describing how “we found our common language with these bracelets” (P1).

Finished objects extended the practice beyond the group. Participants described how exhibitions and shared images sparked curiosity and conversation. Viewers often “instantly think about what data I have” when encountering the pieces (P4). In this way, objects acted less as final representations and more as carriers of shared process and ongoing relations.

5 Discussion

The data crafting collective understood the process of engaging with data as shared, encompassing aspects of care and commitment to the craft and the community around it. During the crafting sessions data passed into the background. The members engaged in an ongoing weaving practice that was manifested materially and conceptually through the sharing of personal stories, creating a bond among the members. Data and making became prompts for sharing knowledge but also personal stories. We introduce data weaving as a long-term, community-based practice that foregrounds process, relationships, and situated engagement with data. Rather than centering finished artifacts, the members of the group emphasized how meaning emerges through recurring gatherings, making, and collective sensemaking over time. In the following, we reflect and position data weaving in relation to previous works that focus on making.

Maker culture research has examined how infrastructures, such as makerspaces shape making practices, not only through technical resources but through social organization and community engagement [13]. Our findings contribute to this work by focusing less on the site of participation, and more on how commitment is sustained through ongoing relationships and shared practice. Data weaving did not emerge through an institutional framing or stated mission. Instead, it took shape gradually through repeated gatherings, open participation, and shared routines. This form of organization differs from many accounts of makerspaces that emphasize spaces, tools, or techniques as the primary conditions for participation. In data weaving, the community itself, and the interest in expressing

data, served as the main organizing structure. While the group sometimes worked toward shared outcomes, such as exhibitions, these goals did not override the informal and voluntary nature of participation. Members balanced collective responsibility with keeping the practice enjoyable, and flexible. This orientation aligns more closely with feminist scholarship on maker and design justice that focuses on community over technical mastery [4]. Thus, the value of the practice lay not only in what was produced, but in the conditions that allowed it to persist over time.

Participants' descriptions of crafting, or rather weaving, as slow, tactile, and resistant to easy correction strongly resonate with how prior work on data physicalization reinforces the idea that handmaking with data redistributes authority away from computational systems toward human judgment [6]. The inability to instantly undo mistakes demanded attention, reshaping how data was encountered and interpreted which we understand as a core principle of the data weaving process. Thus, our findings show that slowness was not only reflective but social. Flow, in particular, was often described as a collective process that was supported by shared silence and parallel activity. This complements the accounts of craft-based research as an embodied way of knowing, where thinking and feeling are entangled in the act of making itself [5]. Related work on data textiles similarly shows that slow, physical data making supports understanding and connection to data [12], and our findings extend this by showing how such understanding is produced collectively through shared making, adding the value of discussion around data. Care emerged indirectly rather than being an explicit goal. Participants did not frame data weaving as therapeutic, yet over time, personal and sensitive topics surfaced through the shared activity of weaving. In fact, craft-based maker cultures have highlighted making as a form of care work that supports wellbeing and social connection [1, 15].

Overall, data weaving shows how collective sensemaking unfolds across months, sustained by regular co-presence and informal learning. Knowledge developed through observation, mutual help, and the gradual formation of a shared material language, echoing accounts of craft ecologies and communities of practice [5]. These position data weaving as a form of collective data practice that foregrounds community as central to data practice. For HCI and design research, this suggests shifting attention away from tools and representations alone, toward the social conditions that allow data practices to be lived and meaningfully shared over time.

6 Conclusion

This paper explored data weaving as a collective practice in which data, materials, and social relations are inseparable. Studying a small, self-organized community of data weavers, we showed that crafting with data is less about representing datasets accurately and more about negotiating meaning through shared making. Participants treated data as something to reflect on together, shaped by conversation, care, and material constraints rather than fixed analytical goals. Our findings extend prior work on data physicalization and data craft by centering community as a key site of sense-making. Instead of focusing on individual reflection or finished artifacts, or the data themselves, we highlight how knowledge emerges through

peer learning and open discussions. Materials and techniques actively shaped interpretation and trust in the data. These insights suggest that data-oriented making should be understood as a social practice. Designing for data craft communities may therefore require supporting openness, shared authorship, and adaptability rather than precision or efficiency. By foregrounding communal making and lived engagement with data, this paper contributes an alternative perspective on how data can be meaningfully worked with, together.

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